

Gaussian Blur

Gaussian filters can be used to blur images or remove high frequency noise.

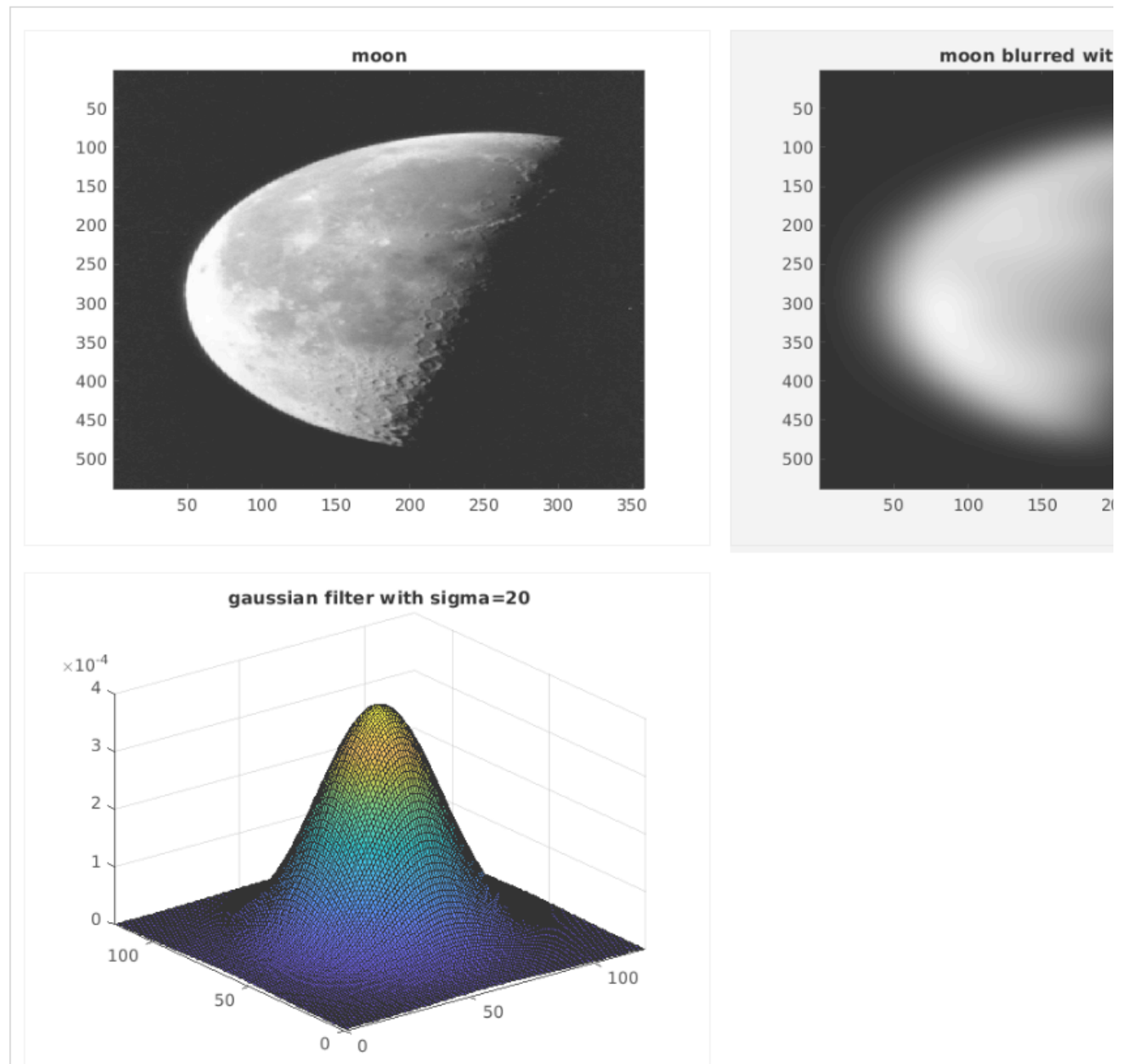
Write a function called **GaussianBlur**, which takes as input an image [MxN] **img** and scalar **sigma**, representing the Gaussian filter sigma. The function returns the Gaussian filtered image, and [WxW] matrix **filt**, the Gaussian filter, as a second output. The function uses 'fspecial' to create **filt** with **W** the next integer.

The steps are:

- Use fspecial to generate the [WxW] matrix filt.
- Apply the filter to the image using convolution.

Running the example code below, your output should look identical to this:

Output



Function ?

[Save](#)[Reset](#)[MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

```
1 function [img_filt,filt] = GaussianBlur(img,sigma)
2
3 % Use fspecial to generate the Gaussian filter
4 w = round(6*sigma);
5 filt = fspecial('gaussian', [w w], sigma);
6 % Apply the filter to the image using convolution
7 img_filt = conv2(img, filt, 'same');
8 end
```

Code to call your function ?

 Reset

```
1 sigma = 20;
2 img = imread('moon.tif');
3 [img_filt,filt] = GaussianBlur(img,sigma);
4 imagesc(img)
5 title('moon')
6 colormap(gray)
7 figure();
8 imagesc(img_filt)
9 title('moon blurred with sigma=20')
10 colormap(gray)
11 figure;
12 surf(filt)
13 title('gaussian filter with sigma=20')
14
```

 Run Function

Assessment: All Tests Passed

Submit

 Is output correct with random valued input